

ELC 2060

Electronic Circuits

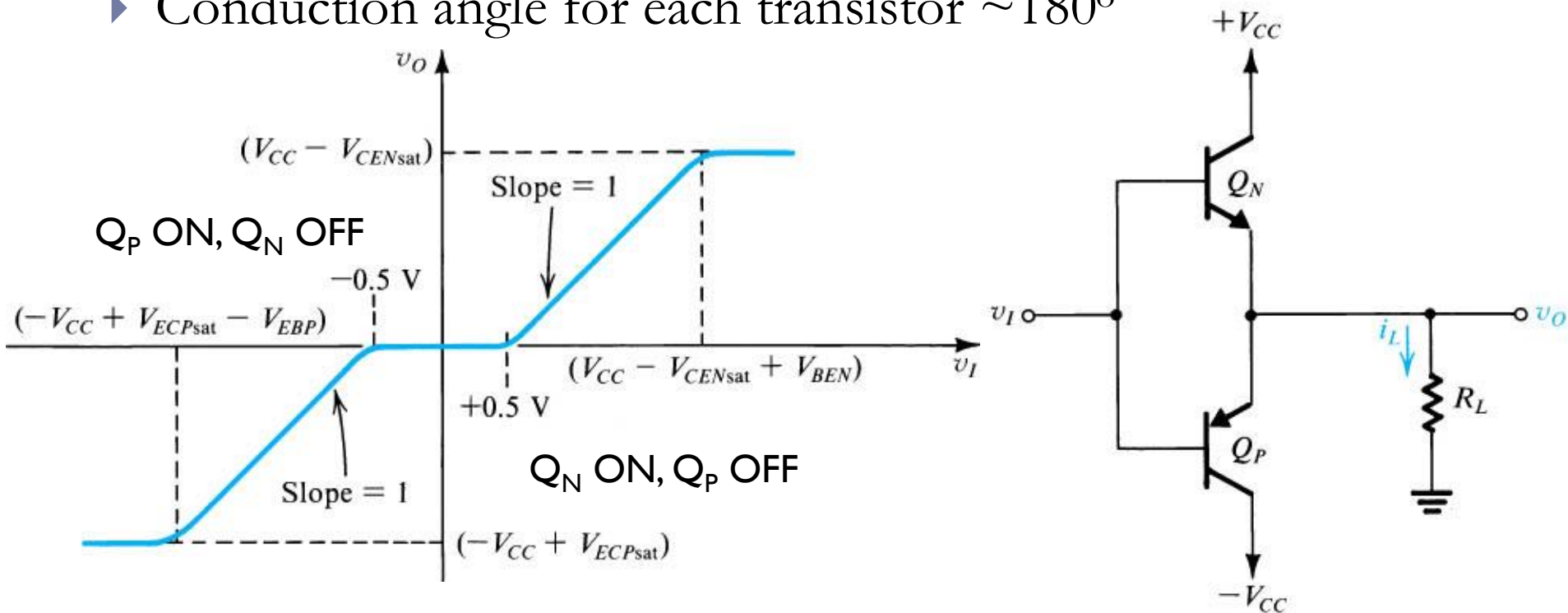
Output Stages and Power Amplifiers

Department of Electronics and Electrical Communications
Engineering

Cairo University

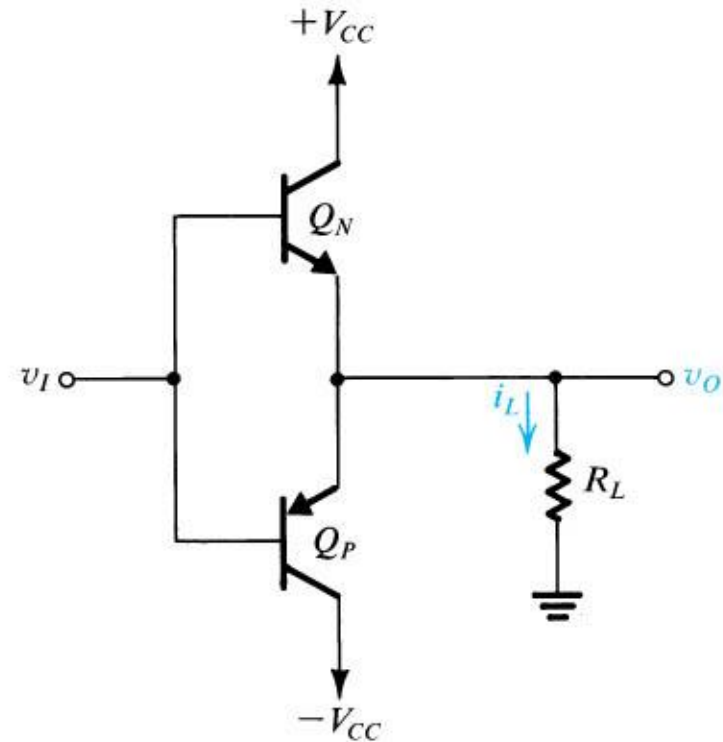
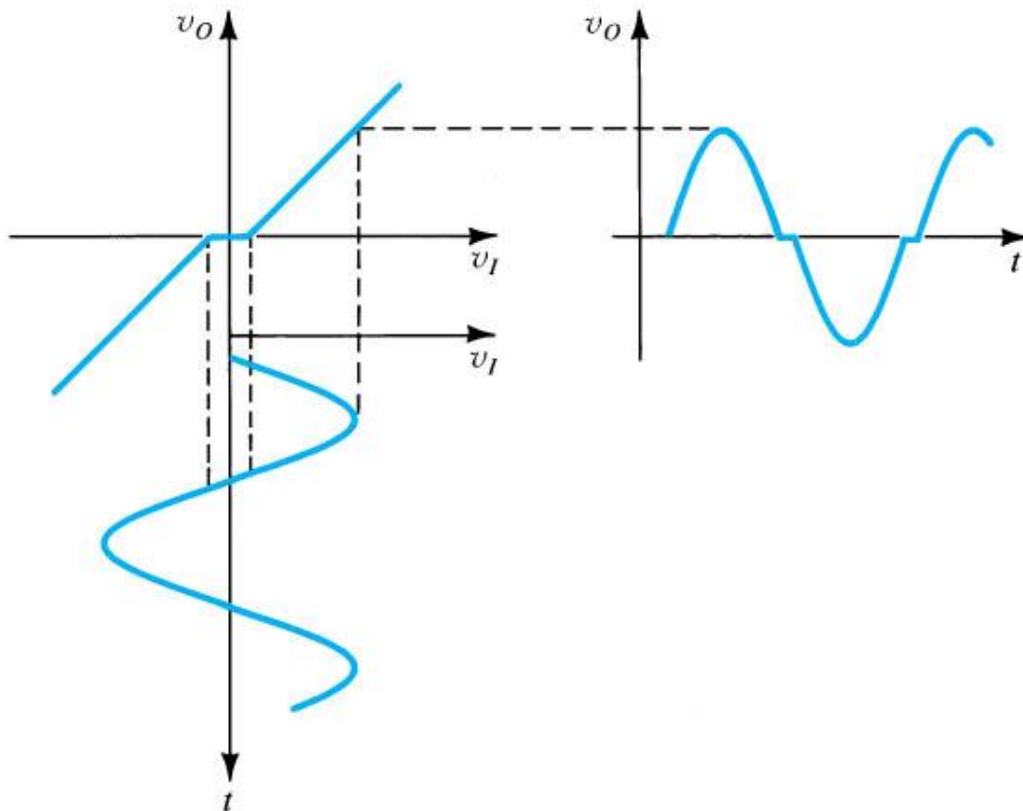
Class B output stage

- ▶ Both transistors are off when the input is zero
 - ▶ Q_N turns on when $v_I > V_{BEN}$, then $v_o = v_I - V_{BEN}$
 - ▶ Q_P turns on when $v_I < -V_{EBP}$, then $v_o = v_I + V_{EBP}$
- ▶ Conduction angle for each transistor $\sim 180^\circ$



Class B output stage

- ▶ There is a region where both transistors are off
 - ▶ Cross over distortion will occur

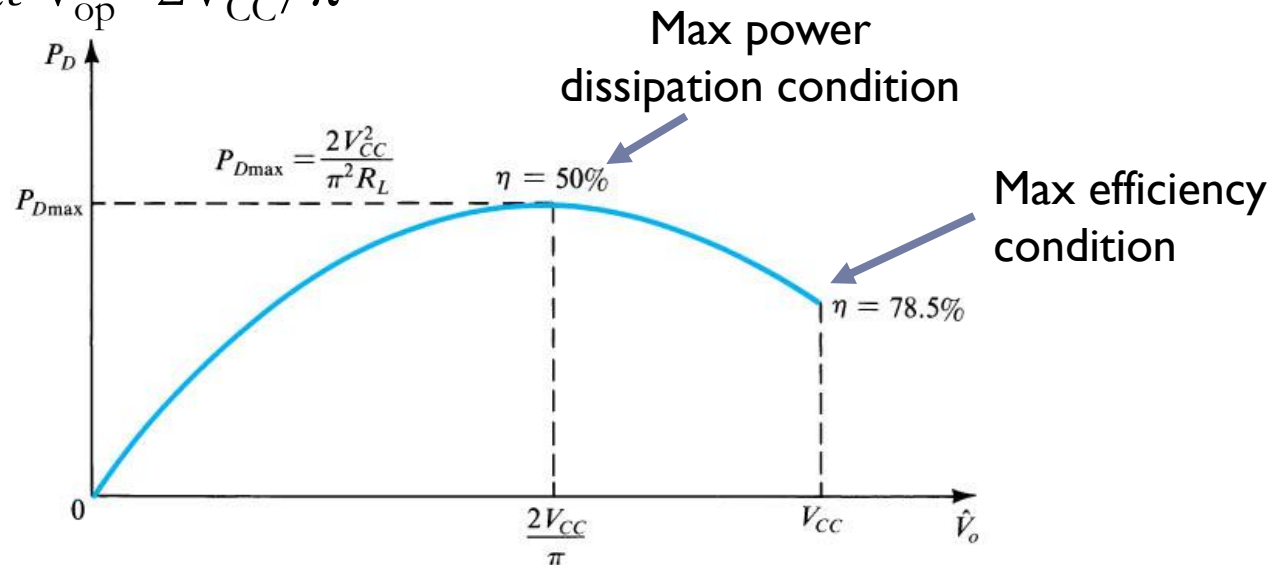


Efficiency of class B output stage

- ▶ Neglecting the cross over distortion
 - ▶ Load power $P_L = V_{op}^2 / 2R_L$
- ▶ The supply current is a half sine wave with amplitude $I_{op} = V_{op} / R_L$
 - ▶ Using Fourier series, the DC average current is $I_{av} = V_{op} / \pi R_L$
- Supply power $P_S = 2I_{av} V_{CC} = 2V_{op} V_{CC} / \pi R_L$
- ▶ $\eta = \frac{P_L}{P_S} = \frac{\pi}{4} \frac{V_{op}}{V_{CC}}$
- ▶ Class B $\eta_{max} = \pi/4 = 78\%$ at $V_{op} = V_{CC}$

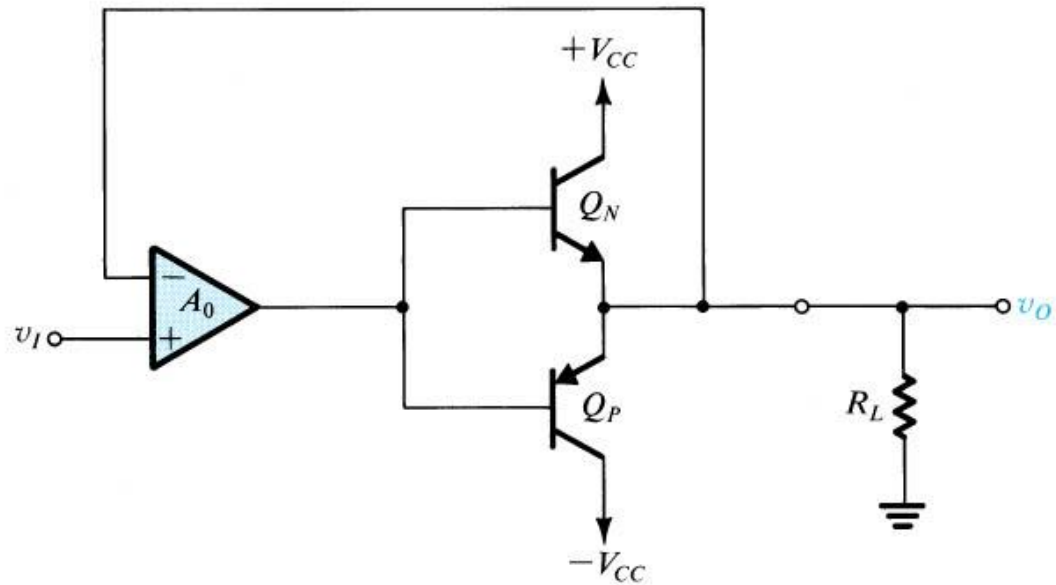
Power dissipation of class B

- ▶ Power dissipated in class B stage is given by
 - ▶ $P_D = P_S - P_L$
 - ▶ $P_D = \frac{2}{\pi} \frac{V_{op}}{R_L} V_{CC} - \frac{1}{2} \frac{V_{op}^2}{R_L}$
 - ▶ By differentiating the previous equation, max power dissipation happens at $V_{op} = 2V_{CC}/\pi$



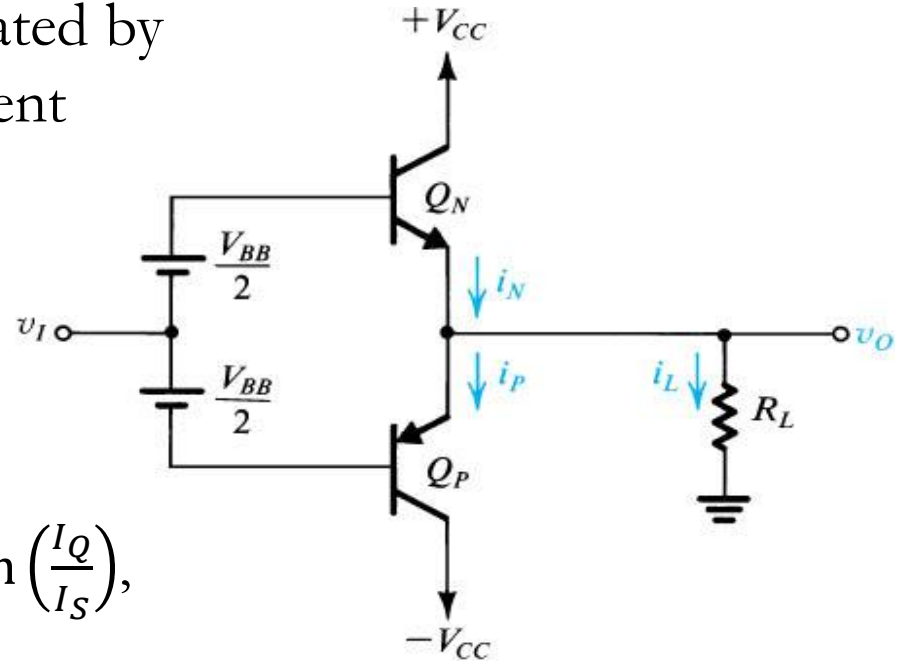
Reducing cross over distortion

- Dead zone is reduced to $\pm V_{BE}/A_0$



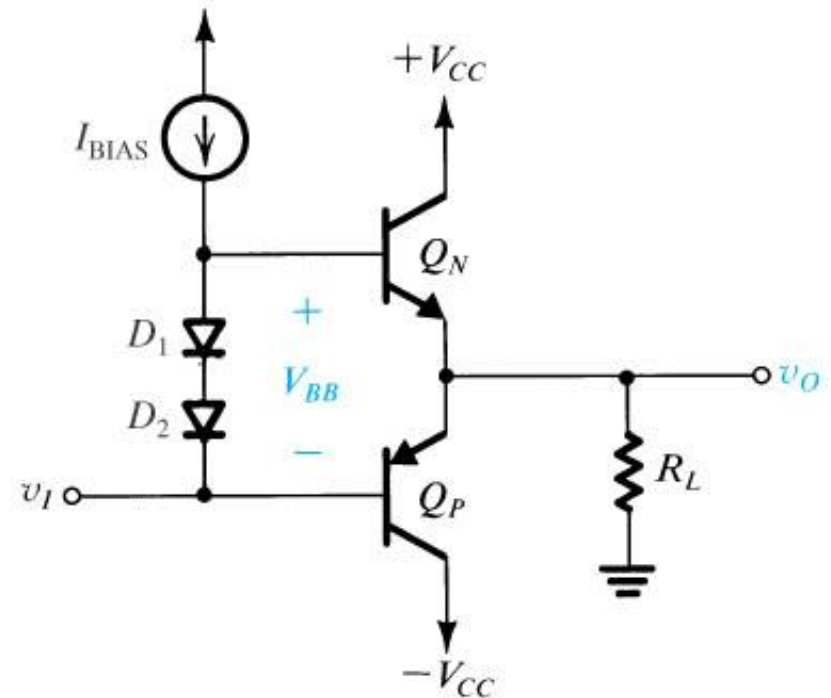
Class AB output stage

- ▶ Cross over distortion can be eliminated by biasing transistor using a small current
- ▶ For $V_{in}=0$ then $V_{out}=0$,
 - ▶ $i_N = i_P = I_Q = I_S e^{V_{BB}/2V_T}$
- ▶ $V_{BEN} + V_{EBP} = V_{BB}$
- ▶ In general
 - ▶ $V_T \ln\left(\frac{i_N}{I_S}\right) + V_T \ln\left(\frac{i_P}{I_S}\right) = V_{BB} = 2V_T \ln\left(\frac{I_Q}{I_S}\right)$,
then $i_N i_P = I_Q^2$ (1)
 - ▶ $i_N = i_P + i_L$ (2)
 - ▶ From (1) and (2) $i_N^2 - i_L i_N - I_Q^2 = 0$



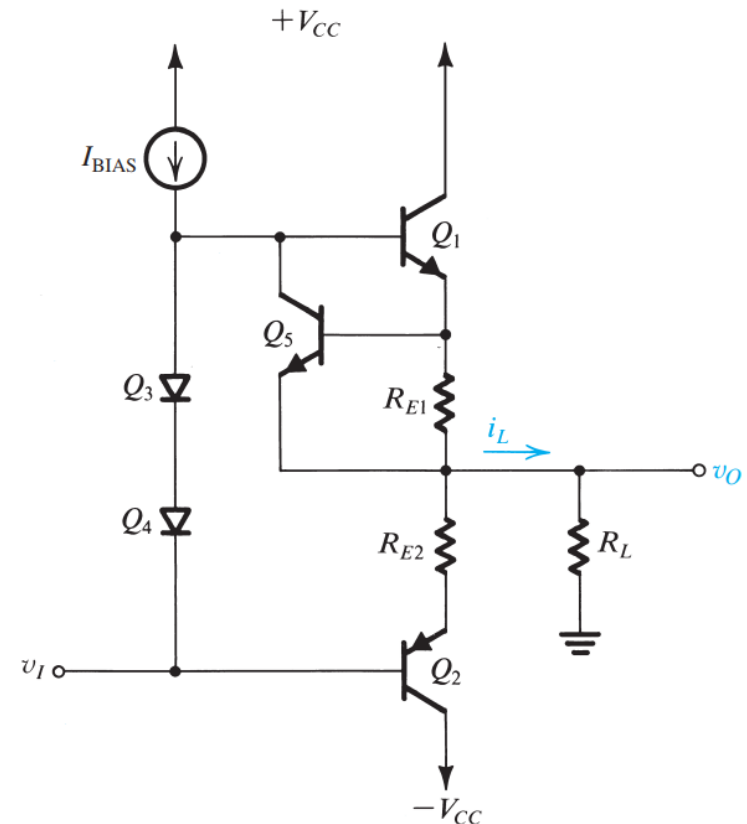
Biasing the class AB amplifier

- ▶ $I_Q = nI_{BIAS}$ where n is the ratio between the emitter area of BJTs and area of diodes
- ▶ Note that I_{BIAS} should be greater than maximum of i_{BN}
 - ▶ $I_{BIAS} > i_{D1,2} + i_{BN}$



Short Circuit Protection for Class AB amplifier

- ▶ For +ve v_o , as R_L decreases-
 - ▶ $I(Q1)$ increases
 - ▶ Voltage drop (R_{E1}) increases
 - ▶ $Q5$ turns on and conducts most of the current
 - ▶ $I(Q1)$ starts to decrease



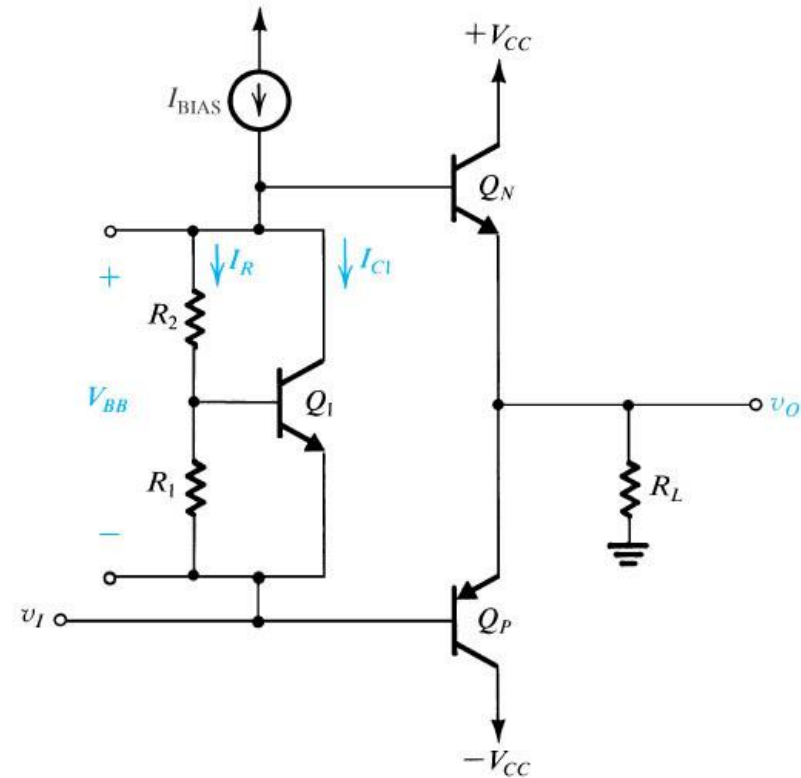
Biasing the class AB amplifier (V_{BE} multiplier)

► Neglecting the base current of Q_1

► $I_R = \frac{V_{BE1}}{R_1}$

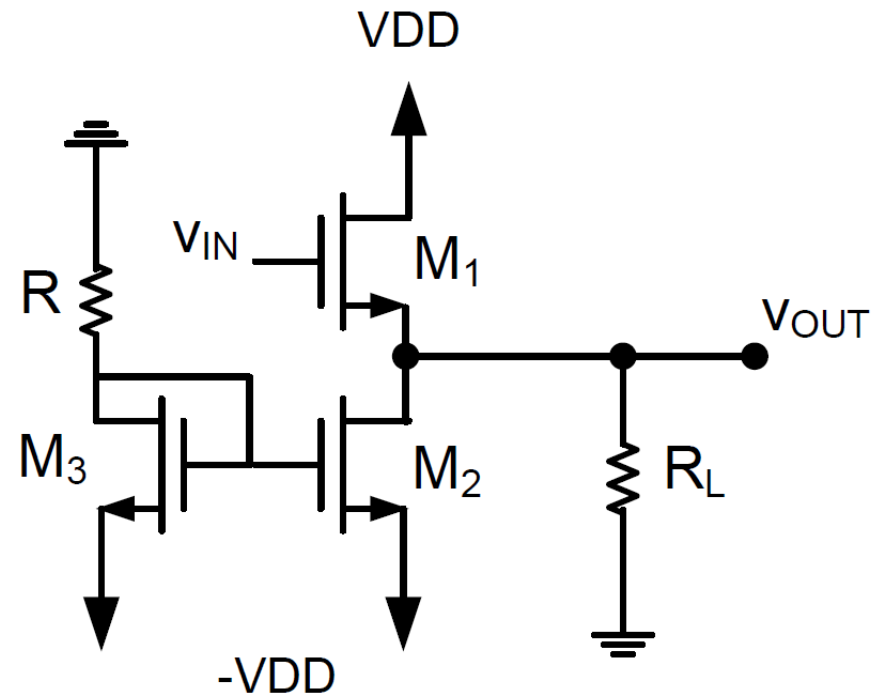
► $V_{BB} = I_R(R_1 + R_2) = V_{BE1} \left(1 + \frac{R_2}{R_1}\right)$

► $V_{BE1} = V_T \ln \left(\frac{I_{C1}}{I_{S1}}\right) = V_T \ln \left(\frac{I_{BIAS} - I_R}{I_{S1}}\right)$



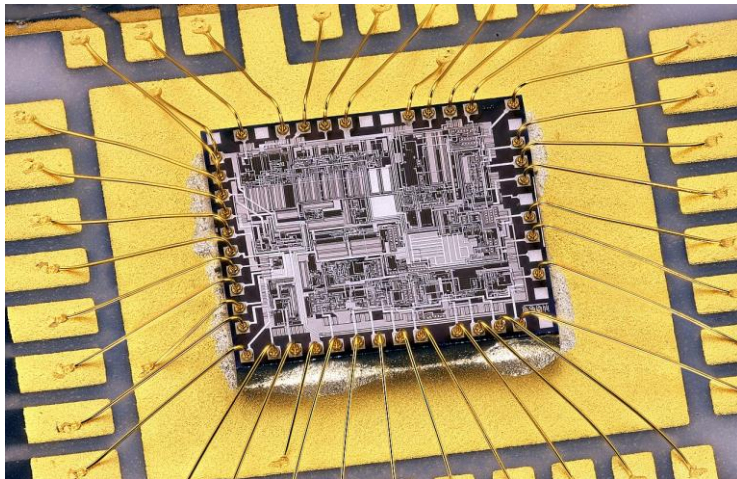
Using MOS in output stages

- ▶ MOS can be used in output stage instead of bipolar
- ▶ Main difference is that V_{ov} of MOS device is a function of current
- ▶ At peak output voltage V_{omax}
 - ▶ $V_{omax} = V_{DD} - V_{ov}$
 - ▶ $V_{ov} = \sqrt{\frac{2(I + V_{omax}/R_L)}{k}}$

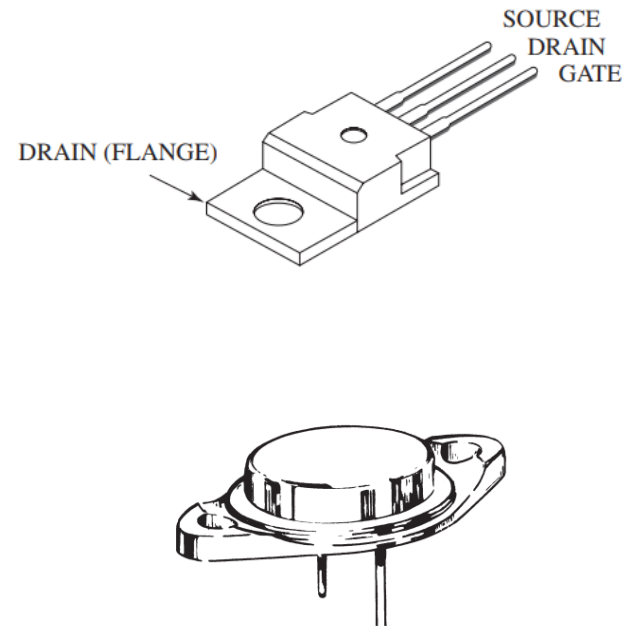


Packaging and Heat Sink

- ▶ Can you deal with bare die?
 - ▶ Wirebond, C4, ...etc
- ▶ Output stage dissipates a lot of heat.
- ▶ How can we deal with that?
 - ▶ Use heat sink



https://en.wikipedia.org/wiki/Wire_bonding#/media/File:07R01.jpg

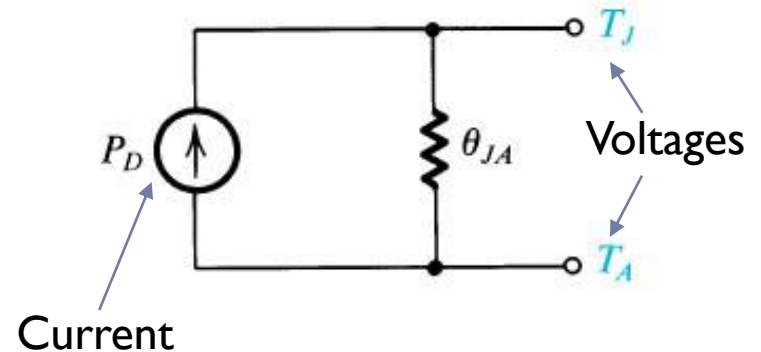


Thermal Resistance

- ▶ Power transistors dissipate large power $\rightarrow$ converted into heat
- ▶ The temperature rise is proportional with the power consumption:

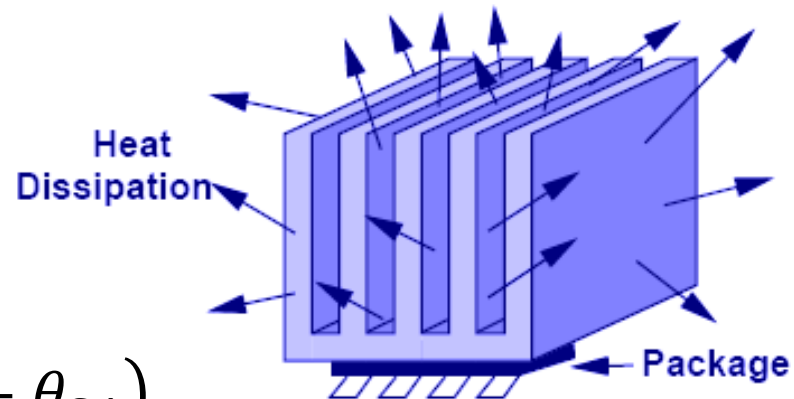
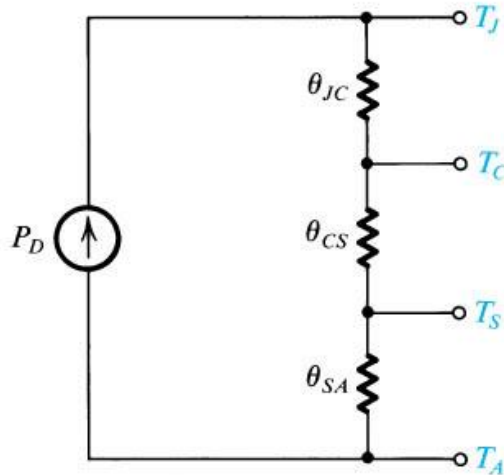
$$T_J - T_A = \theta_{JA} P_D$$

- T_A is the ambient temperature, T_J is the junction temperature
- T_J shouldn't exceed $T_{J_{\max}}$ ($\sim 125^\circ\text{C}$ for Si)
- θ_{JA} is the thermal resistance between junction & ambient ($^\circ\text{C}/\text{W}$)
- Thermal conduction is analogous to electrical conduction



Thermal Resistance (Cont'd)

- ▶ To dissipate the heat out of the circuit, a heat sink is used



$$T_J - T_A = P_D(\theta_{JC} + \theta_{CS} + \theta_{SA})$$

- ▶ θ_{JC} : **j**unction to **c**ase thermal resistance
- ▶ θ_{CS} : **c**ase to heat **s**ink thermal resistance
- ▶ θ_{SA} : heat **s**ink to **a**mbient thermal resistance